Theoretical Appendix

What Do Time Series Foundation Models Actually Learn?
A Geometric Perspective

Contents

Overview and Notation

This appendix provides rigorous theoretical foundations for the Latent Geometry Framework (LGF) introduced in the main paper. Each section treats one component of LGF: the existence and structure of the Temporal Representation Manifold (TRM); the formal properties of the Latent Trajectory Complexity (LTC); stability theory underlying the TSI and GSP metrics; information-theoretic guarantees for the Class Separation Score (CSS); the geometric basis for the reliability metrics PDS and FSS; and generalization bounds derived from manifold complexity.

Throughout, we use the following notation consistently.

$f_\theta : \mathbb{R}^L \to \mathbb{R}^D$	Frozen TSFM encoder
$z_i = f_\theta(w_i) \in \mathbb{R}^D$	Embedding of window w_i
$\mathcal{Z} = \{z_i\}_{i=1}^N$	Empirical embedding set
$\mathcal{M}_\theta = f_\theta(\mathrm{supp}(\mathcal{P}))$	Temporal Representation Manifold
L_θ	Lipschitz constant of f_θ
d_{int}	Intrinsic dimension of $\mathcal{M}_\theta$
$e_i \in \mathbb{R}$	Forecast error for window w_i
$\bar{d}_i$	Mean k-NN distance of z_i in $\mathcal{Z}$
$y_i^c \in \{0, 1\}$	Binary label for temporal concept c
$J_{f_\theta}(w)$	Jacobian matrix of f_θ at w
$\sigma_{\max}(A)$	Largest singular value of matrix A
$H_b(p)$	Binary entropy $-p \log p - (1-p)\log(1-p)$
$I(X; Y)$	Mutual information between X and Y
$S^{D-1}(r)$	Sphere of radius r in $\mathbb{R}^D$

1 Geometric Foundations of the Temporal Representation Manifold

1.1 Existence and Rectifiability

The central object of our analysis is the image of the encoder applied to the support of the window distribution.

Definition 1.1 (Temporal Representation Manifold)**.** Let $\mathcal{P}$ be the distribution over context windows $w \in \mathbb{R}^L$ drawn from a target time series. The *Temporal Representation Manifold* is

$$\mathcal{M}_\theta \;=\; f_\theta\big(\mathrm{supp}(\mathcal{P})\big) \;\subset\; \mathbb{R}^D.$$

We place minimal assumptions on the encoder.

Assumption 1.2 (Lipschitz Encoder)**.** The encoder f_θ is L_θ-Lipschitz: there exists $L_\theta < \infty$ such that for all $w, w' \in \mathbb{R}^L$,

$$\|f_\theta(w) - f_\theta(w')\|_2 \;\le\; L_\theta \;\|w - w'\|_2 \,.$$

Assumption 1.3 (Compact Support)**.** The support $\mathrm{supp}(\mathcal{P}) \subset \mathbb{R}^L$ is compact.

Transformer encoders satisfy Assumption 1.2 when activation functions are Lipschitz (ReLU has Lip = 1; GELU is locally Lipschitz with bounded derivative); Assumption 1.3 holds whenever window values are bounded, which is enforced by window-level normalisation in our experimental protocol.

Theorem 1.4 (TRM Compactness and Rectifiability)**.** *Under Assumptions 1.2 and 1.3, the TRM $\mathcal{M}_\theta$ is compact and rectifiable: it has finite d-dimensional Hausdorff measure for some $d \le L$, and can be covered by finitely many Lipschitz images of bounded subsets of $\mathbb{R}^d$.*

Proof. Compactness: $\mathrm{supp}(\mathcal{P})$ is compact by Assumption 1.3, and the continuous image of a compact set is compact, so $\mathcal{M}_\theta$ is compact.

Rectifiability: By Rademacher's theorem, a Lipschitz map $f_\theta : \mathbb{R}^L \to \mathbb{R}^D$ is differentiable almost everywhere (a.e.) with respect to Lebesgue measure. The image $\mathcal{M}_\theta$ is therefore a Lipschitz image of $\mathrm{supp}(\mathcal{P})$. By Federer's structure theorem for rectifiable sets [Federer, 1969], the image of a compact set under a Lipschitz map from $\mathbb{R}^L$ is countably m-rectifiable for $m \le L$: it can be covered (up to a null set) by countably many Lipschitz images of bounded subsets of $\mathbb{R}^m$. Since $\mathcal{M}_\theta$ is compact, a finite sub-cover suffices. $\square$

Theorem 1.5 (Intrinsic Dimension Bound)**.** *Let f_θ be continuously differentiable. For $\mathcal{P}$-almost every window w, the local dimension of $\mathcal{M}_\theta$ at $z = f_\theta(w)$ is bounded by the rank of the Jacobian:*

$$d_{\mathrm{int}}(z) \;\le\; \mathrm{rank}\big(J_{f_\theta}(w)\big) \;\le\; \min(L, D).$$

Proof. By the rank theorem in differential geometry, the image of the differential $\mathrm{d}f_\theta|_w : T_w\mathbb{R}^L \to T_z\mathbb{R}^D$ has dimension $\mathrm{rank}(J_{f_\theta}(w))$. The tangent space of $\mathcal{M}_\theta$ at z is contained in this image, so the local intrinsic dimension of $\mathcal{M}_\theta$ at z is at most $\mathrm{rank}(J_{f_\theta}(w))$. The bound $\mathrm{rank}(J_{f_\theta}(w)) \le \min(L, D)$ follows from the definition of rank. $\square$

Remark 1.6. Theorem 1.5 connects directly to the empirical observation that pretrained TSFMs produce more compact representations than per-dataset supervised Transformers. A pretrained encoder operating across many domains is expected to produce a lower-rank Jacobian on any single domain—it does not "expand" the representation to fit domain-specific fluctuations—which explains the $3.5\times$ lower LTC step relative to the supervised Transformer baseline.

1.2 Concentration of Measure in High Dimensions

A central challenge in analyzing high-dimensional embedding spaces is that pairwise distances concentrate strongly, potentially masking any semantic structure. The following results quantify this precisely and motivate the pretraining comparison in LGF.

Theorem 1.7 (Concentration of Pairwise Distances)**.** *Let $z_1, \ldots, z_N$ be drawn i.i.d. from a spherically symmetric distribution on $S^{D-1}(\sqrt{D})$ (the sphere of radius $\sqrt{D}$ in $\mathbb{R}^D$). For any $\epsilon > 0$ and any pair $i \neq j$,*

$$\mathbb{P}\Big(\Big|\|z_i - z_j\|_2 - \sqrt{2D}\Big| > \epsilon\sqrt{D}\Big) \;\leq\; 2\exp\big(-c_0\,\epsilon^2 D\big)\,,$$

for an absolute constant $c_0 > 0$.

Proof. Define $g : S^{D-1}(\sqrt{D}) \to \mathbb{R}$ by $g(z) = \|z - z_j\|_2$, which is 1-Lipschitz for any fixed z_j. By Lévy's lemma for the sphere (see, e.g., Ledoux 2001), a 1-Lipschitz function on $S^{D-1}(\sqrt{D})$ satisfies

$$\mathbb{P}\big(|g(z) - M_g| > t\big) \;\leq\; 2\exp\Big(-\tfrac{(D-1)\,t^2}{2D}\Big)$$

where M_g is its median. For spherically symmetric distributions the expected distance is $\mathbb{E}[\|z_i - z_j\|_2] = \sqrt{2D}(1+O(1/D))$, and the median and mean agree to within $O(1/\sqrt{D})$. Setting $t = \epsilon\sqrt{D}$ gives the stated bound. □

Corollary 1.8 (Silhouette Degeneracy for Random Embeddings)**.** *Let $z_1, \ldots, z_N$ be drawn i.i.d. from a spherically symmetric distribution on $S^{D-1}(\sqrt{D})$, and let $\mathbf{y}$ be any labeling. Then the expected LCS satisfies*

$$\mathbb{E}\big[\mathrm{LCS}(\mathcal{Z}, \mathbf{y})\big] \;\to\; 0 \quad \textit{as } D \to \infty.$$

Proof. The silhouette score for point i is $(b(i) - a(i))/\max(a(i), b(i))$, where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance. By Theorem 1.7, all pairwise distances concentrate around $\sqrt{2D}$: for any $\epsilon > 0$, $|a(i) - \sqrt{2D}| \leq \epsilon\sqrt{D}$ and $|b(i) - \sqrt{2D}| \leq \epsilon\sqrt{D}$ with probability at least $1 - 2N^2\exp(-c_0\epsilon^2 D) \to 1$ as $D \to \infty$ (by union bound, provided $\log N = o(D)$). Therefore $b(i) - a(i) = O(\epsilon\sqrt{D})$ while $\max(a(i), b(i)) = \Theta(\sqrt{D})$, giving silhouette score $O(\epsilon)$ for any fixed ϵ. Since ϵ was arbitrary, $\mathbb{E}[\mathrm{LCS}] \to 0$. □

Remark 1.9. Corollary 1.8 explains why LCS values are small even for pretrained models in high-dimensional spaces ($D = 512$ for TimesFM). It establishes that the pretrained–vs–random LCS gap, rather than the absolute LCS value, is the meaningful signal. It also motivates using linear probing (CSS) as the primary cluster-organisation diagnostic, since linear probes can detect class-mean separation even when silhouette scores are near zero.

2 Latent Trajectory Complexity: Formal Properties

2.1 LTC as an Arc Length Approximation

Let σ be the permutation sorting windows by start time s_i, and define temporally adjacent displacement vectors $\delta_i = z_{\sigma(i+1)} - z_{\sigma(i)} \in \mathbb{R}^D$.

Definition 2.1 (Temporal Trajectory)**.** The *temporal trajectory* $\Gamma_N : [0, N-1] \to \mathbb{R}^D$ is the piecewise linear interpolation through the time-ordered embeddings: $\Gamma_N(t) = (1 - \{t\})\, z_{\sigma(\lfloor t \rfloor+1)} + \{t\}\, z_{\sigma(\lfloor t \rfloor+2)}$, where $\{t\}$ denotes the fractional part.

Proposition 2.2 ($\mathrm{LTC}_{\mathrm{step}}$ as Normalised Arc Length)**.** $\mathrm{LTC}_{\mathrm{step}} = \mathrm{Length}(\Gamma_N)\,/\,(N-1)$, *where* $\mathrm{Length}(\Gamma_N) = \sum_{i=1}^{N-1} \|\delta_i\|_2$ *is the total arc length of the piecewise linear trajectory.*

Proof. By Definition 2.1, Γ_N is piecewise linear with segments $\delta_i = z_{\sigma(i+1)} - z_{\sigma(i)}$. The arc length of a piecewise linear curve is the sum of segment lengths: $\mathrm{Length}(\Gamma_N) = \sum_{i=1}^{N-1} \|\delta_i\|_2$. Dividing by $N-1$ gives $\mathrm{LTC}_{\mathrm{step}}$ as defined. □

Theorem 2.3 ($\mathrm{LTC}_{\mathrm{dir}}$ Characterises Directional Consistency)**.** $\mathrm{LTC}_{\mathrm{dir}} \in [-1, 1]$ *with:*

(i) $\mathrm{LTC}_{\mathrm{dir}} = 1$ *if and only if all consecutive displacement vectors* δ_i, δ_{i+1} *are positively proportional (*$\delta_i = \lambda_i \delta_{i+1}$*,* $\lambda_i > 0$ *for all* i*), i.e., the trajectory is a straight ray.*

(ii) $\mathrm{LTC}_{\mathrm{dir}} = -1$ *if and only if all consecutive displacement vectors are anti-parallel (*$\lambda_i < 0$ *for all* i*), i.e., the trajectory alternates direction along a line.*

(iii) $\mathrm{LTC}_{\mathrm{dir}} = 0$ *if consecutive steps are uncorrelated in direction, corresponding to a random walk in embedding space.*

Proof. Each term $\cos\theta_i = \delta_i \cdot \delta_{i+1} / (\|\delta_i\|\,\|\delta_{i+1}\|)$ is a cosine similarity, hence bounded in $[-1, 1]$. The average of values in $[-1, 1]$ lies in $[-1, 1]$.

(i): By the Cauchy-Schwarz equality condition, $\cos\theta_i = 1$ iff δ_i and δ_{i+1} are positively proportional. Averaging: $\mathrm{LTC}_{\mathrm{dir}} = 1$ iff all $\cos\theta_i = 1$, i.e., all consecutive steps point in the same direction.

(ii): Similarly, $\cos\theta_i = -1$ iff δ_i and δ_{i+1} are anti-parallel. If all terms equal -1, the trajectory oscillates along a fixed line.

(iii): If δ_i are i.i.d. uniform on S^{D-1}, then $\mathbb{E}[\cos\theta_i] = 0$ for $D \geq 2$ (since $\mathbb{E}[\delta_i \cdot \delta_{i+1}] = \mathbb{E}[\delta_i] \cdot \mathbb{E}[\delta_{i+1}] = 0$ by independence and spherical symmetry). By the law of large numbers, $\mathrm{LTC}_{\mathrm{dir}} \to 0$ as $N \to \infty$. □

Remark 2.4. Theorem 2.3 establishes that $\mathrm{LTC}_{\mathrm{dir}} \approx -0.4$ for all models (pretrained and random) in our experiments is consistent with *locally oscillatory* trajectories. This is expected for time-series embeddings: adjacent windows share a large overlapping segment (stride 64 within window length 512), so their embeddings often move in complementary directions as the leading and trailing data change.

2.2 Pretraining Amplifies Trajectory Complexity

We now provide a theoretical basis for the empirically observed gap $\mathrm{LTC}_{\mathrm{step}}(\text{pretrained}) > \mathrm{LTC}_{\mathrm{step}}(\text{random})$.

Assumption 2.5 (Non-Degenerate Temporal Variation)**.** The time series has non-trivial temporal autocorrelation: adjacent windows w_t, w_{t+1} differ in at least one principal direction with $\mathbb{E}[\|w_{t+1} - w_t\|_2^2] = \sigma_w^2 > 0$.

Theorem 2.6 (Pretraining Amplifies $\mathrm{LTC}_{\mathrm{step}}$)**.** *Under Assumption 2.5, let f_θ be a pretrained encoder and f_{θ_0} be the same architecture with random initialisation, with both satisfying Assumption 1.2. If the pretraining objective $\mathcal{L}(\theta)$ requires the encoder to be sensitive to temporal context—i.e., $\nabla_\theta \mathcal{L}(\theta)$ is non-degenerate and increases the directional sensitivity of f_θ to input differences—then:*

$$\mathbb{E}[\mathrm{LTC}_{\mathrm{step}}(f_\theta)] \;\geq\; \mathbb{E}[\mathrm{LTC}_{\mathrm{step}}(f_{\theta_0})]\,.$$

Equality holds only if pretraining leaves the encoder indifferent to temporal variation.

Proof. By Proposition 2.2:

$$\mathbb{E}[\mathrm{LTC}_{\mathrm{step}}(f_\theta)] = \frac{1}{N-1}\sum_{i=1}^{N-1} \mathbb{E}\big[\big\|f_\theta(w_{\sigma(i+1)}) - f_\theta(w_{\sigma(i)})\big\|_2\big]\,.$$

For any encoder f, by the Cauchy-Schwarz inequality:

$$\mathbb{E}[\|f(w') - f(w)\|_2] \;\geq\; \frac{|\langle v^*,\;\; \mathbb{E}[f(w') - f(w)]\;\rangle|}{\|v^*\|_2}$$

for any unit vector $v^* \in \mathbb{R}^D$. The pretraining objective trains f_θ to produce representations that vary with temporal context: specifically, an autoregressive or masked prediction objective requires that $f_\theta(w_{t+1}) \neq f_\theta(w_t)$ in expectation (otherwise the encoder outputs a constant and provides no useful representation for prediction). This means the expected displacement $\mathbb{E}[f_\theta(w') - f_\theta(w)]$ along the direction most sensitive to temporal change is strictly positive for pretrained f_θ and zero for random f_{θ_0} (whose parameters are initialised independently of the data). Therefore $\mathbb{E}[\mathrm{LTC}_{\mathrm{step}}(f_\theta)] \geq \mathbb{E}[\mathrm{LTC}_{\mathrm{step}}(f_{\theta_0})]$, with equality only if pretraining provides no update in the temporal-sensitivity direction. □

3 Lipschitz Stability and the TSI

3.1 Lipschitz Constants of Transformer Encoders

We first establish how the Lipschitz constant of a Transformer encoder decomposes over its constituent operations.

Lemma 3.1 (Residual Connection Lipschitz Bound)**.** *Let $F : \mathbb{R}^D \to \mathbb{R}^D$ be L_F-Lipschitz. Then the residual map $h(x) = x + F(x)$ is $(1 + L_F)$-Lipschitz.*

Proof. For any $x, y \in \mathbb{R}^D$:

$$\|h(x) - h(y)\|_2 = \|(x-y) + (F(x) - F(y))\|_2 \leq \|x-y\|_2 + \|F(x) - F(y)\|_2 \leq (1+L_F)\,\|x-y\|_2\,.$$

□

Lemma 3.2 (Multi-Head Self-Attention Lipschitz Bound)**.** *Let* MSA *be a multi-head self-attention layer with query/key/value projection matrices* $W_Q, W_K, W_V \in \mathbb{R}^{D\times D}$ *and output projection* $W_O \in \mathbb{R}^{D\times D}$*. If the softmax attention weights are treated as fixed (frozen at inference), then:*

$$\mathrm{Lip}(\mathrm{MSA}) \;\le\; \sigma_{\max}(W_O)\,\sigma_{\max}(W_V).$$

Proof. In the frozen-weight regime the attention weight matrix $A \in \mathbb{R}^{T\times T}$ (where T is the sequence length) is a fixed row-stochastic matrix, so $\|A\|_2 \le 1$. The output is $\mathrm{MSA}(X) = A\,XW_VW_O$. For input perturbation ΔX:

$$\|A\,\Delta X W_V W_O\|_F \le \|A\|_2\,\|\Delta X\|_F\,\sigma_{\max}(W_V)\sigma_{\max}(W_O) \le \|\Delta X\|_F\;\sigma_{\max}(W_V)\sigma_{\max}(W_O).$$

□

Theorem 3.3 (Transformer Encoder Lipschitz Constant)**.** *An* $\mathcal{L}$*-layer Transformer encoder* f_θ*, where each layer* ℓ *consists of multi-head self-attention (MSA*$^\ell$*) and a feed-forward network (FFN*$^\ell$ $= W_2^\ell \circ \phi \circ W_1^\ell$ *with* 1*-Lipschitz activation* ϕ*), has Lipschitz constant at most*

$$L_\theta \;\le\; \prod_{\ell=1}^{\mathcal{L}}\bigl(1+\sigma_{\max}(W_O^\ell)\sigma_{\max}(W_V^\ell)\bigr)\bigl(1+\sigma_{\max}(W_2^\ell)\sigma_{\max}(W_1^\ell)\bigr).$$

Proof. By induction on ℓ. For a single layer:

(i) The residual self-attention sub-block $h_1(x) = x+\mathrm{MSA}^\ell(x)$ is $(1+\sigma_{\max}(W_O^\ell)\sigma_{\max}(W_V^\ell))$-Lipschitz by Lemmas 3.1 and 3.2.

(ii) The FFN is $(\sigma_{\max}(W_2^\ell)\sigma_{\max}(W_1^\ell))$-Lipschitz (as a composition of two linear maps and a 1-Lipschitz activation).

(iii) The residual FFN sub-block $h_2(x) = x + \mathrm{FFN}^\ell(x)$ is $(1+\sigma_{\max}(W_2^\ell)\sigma_{\max}(W_1^\ell))$-Lipschitz by Lemma 3.1.

(iv) The full layer $h = h_2 \circ h_1$ has Lipschitz constant at most the product of those of h_1 and h_2.

The encoder f_θ is the composition of $\mathcal{L}$ such layers (plus the embedding and pooling steps, which are linear and hence $O(1)$-Lipschitz). The Lipschitz constant of a composition is bounded by the product of the per-layer constants. □

Remark 3.4. For TimesFM 2.0 ($\mathcal{L} = 50$ layers, $D = 1024$), the bound in Theorem 3.3 can be large in the worst case, but *spectral norm regularisation* during pretraining (implicit via weight decay and gradient clipping) keeps individual spectral norms close to 1. Empirically, the near-unity TSI values (≈0.999) observed in our experiments are consistent with L_θ being small relative to the scale of natural perturbations.

3.2 TSI Lower Bound

Theorem 3.5 (TSI Lower Bound under Lipschitz Continuity). *Let f_θ have Lipschitz constant L_θ, and let $\mathcal{P}$ be a perturbation operator with $\|\mathcal{P}(w) - w\|_2 \le \epsilon$ almost surely. Denote $\tilde{z}_i = f_\theta(\mathcal{P}(w_i))$ and $z_i = f_\theta(w_i)$, and assume $\|z_i\|_2 \ge \mu > 0$. Then the TSI satisfies:*

$$\mathrm{TSI}_{\mathcal{P}} \;\ge\; \frac{\mu - L_\theta \epsilon}{\mu + L_\theta \epsilon},$$

provided $L_\theta \epsilon < \mu$.

Proof. By Lipschitz continuity, $\|\tilde{z}_i - z_i\|_2 \le L_\theta \epsilon$. Write $\tilde{z}_i = z_i + \Delta_i$ with $\|\Delta_i\|_2 \le L_\theta \epsilon$. The cosine similarity for window i is:

$$\frac{z_i \cdot \tilde{z}_i}{\|z_i\|\,\|\tilde{z}_i\|} = \frac{\|z_i\|^2 + z_i \cdot \Delta_i}{\|z_i\|\;\|z_i + \Delta_i\|}.$$

Lower bounding the numerator by Cauchy-Schwarz: $z_i \cdot \Delta_i \ge -\|z_i\|\;\|\Delta_i\| \ge -\mu\, L_\theta \epsilon$. Upper bounding the denominator by the triangle inequality: $\|z_i + \Delta_i\| \le \|z_i\| + \|\Delta_i\| \le \mu + L_\theta \epsilon$. Therefore:

$$\frac{z_i \cdot \tilde{z}_i}{\|z_i\|\,\|\tilde{z}_i\|} \;\ge\; \frac{\mu^2 - \mu\, L_\theta \epsilon}{\mu(\mu + L_\theta \epsilon)} = \frac{\mu - L_\theta \epsilon}{\mu + L_\theta \epsilon}.$$

Averaging over $i = 1, \ldots, N$ preserves the inequality. □

Corollary 3.6 (TSI Approaches Unity for Small Perturbations). *As $\epsilon \to 0$, $\mathrm{TSI}_{\mathcal{P}} \to 1$.*

Proof. Direct substitution in Theorem 3.5: the ratio $(\mu - L_\theta \epsilon)/(\mu + L_\theta \epsilon) \to 1$ as $\epsilon \to 0$. □

3.3 GSP Bounds from TSI

Theorem 3.7 (GSP Controlled by TSI Deficit). *Let $\bar{d} = N^{-2} \sum_{i,j} \|z_i - z_j\|_2 > 0$ be the mean pairwise embedding distance. If $\mathrm{TSI}_{\mathcal{P}} \ge 1 - \kappa$ for some $\kappa \in [0, 1]$, and if all embeddings satisfy $\|z_i\|_2 \le R$, then:*

$$|\mathrm{GSP}_{\mathcal{P}}| \;=\; |\mathrm{LCS}_{\mathrm{clean}} - \mathrm{LCS}_{\mathcal{P}}| \;\le\; \frac{4R\,\kappa^{1/2}}{\bar{d}}.$$

Proof. Denote $\Delta_i = \tilde{z}_i - z_i$. Since $\mathrm{TSI}_{\mathcal{P}} \ge 1 - \kappa$, the average cosine similarity is at least $1 - \kappa$, which implies $\mathbb{E}_i[\|\Delta_i\|_2] \le 2R\kappa^{1/2}$ (by a standard angle-to-distance bound: $\cos\theta \ge 1 - \kappa$ implies the angle $\theta \le \arccos(1 - \kappa) \approx \sqrt{2\kappa}$ for small κ, and $\|\Delta\| = 2R\sin(\theta/2) \le R\theta \le 2R\kappa^{1/2}$).

The LCS is a function of pairwise distances. The intra-cluster distance $a(i)$ and nearest-cluster distance $b(i)$ are each perturbed by at most $\|\Delta_i\|_2 + \max_j \|\Delta_j\|_2 \le 4R\kappa^{1/2}$ in the worst case. Therefore the silhouette score for each point changes by at most $4R\kappa^{1/2}/\bar{d}$ (since $\max(a(i), b(i)) \ge \bar{d}/2$ in a non-degenerate distribution), and the mean LCS changes by the same amount. □

Remark 3.8. Theorem 3.7 shows that models with high TSI (near 1, i.e., $\kappa \approx 0$) also have small GSP, consistent with our empirical finding that GSP ≈ 0 whenever TSI ≈ 1.0. The bound is tight in the sense that it becomes vacuous precisely when TSI $= 0$.

4 Linear Probing Theory: CSS Guarantees

4.1 CSS and Fisher's Linear Discriminant

We analyze CSS through the lens of Fisher's linear discriminant analysis, which provides the geometric interpretation of logistic regression in balanced binary classification.

Definition 4.1 (Fisher Discriminant Ratio)**.** For embeddings $\{z_i\}$ with binary labels $\{y_i^c\} \in \{0, 1\}$, let $\mu_+ = \mathbb{E}[z|y^c = 1]$, $\mu_- = \mathbb{E}[z|y^c = 0]$, and Σ be the pooled within-class covariance. The *Fisher discriminant ratio* is:

$$J_F \; = \; (\mu_+ - \mu_-)^\top \Sigma^{-1} (\mu_+ - \mu_-).$$

Theorem 4.2 (CSS Exceeds Chance iff Classes Are Linearly Separable)**.** CSS$(c) > 1/2$ *if and only if the class means μ_+ and μ_- are distinct in the Mahalanobis metric, i.e., $J_F > 0$.*

Proof. The balanced accuracy of an optimal linear classifier (which logistic regression approximates in the large-sample limit) equals $\Phi(\sqrt{J_F/4})$, where Φ is the standard normal CDF and we have used the Gaussian approximation for linear projections (a consequence of the central limit theorem for bounded embeddings). Since $\Phi(0) = 1/2$, we have:

$$\mathrm{CSS}(c) > 1/2 \iff \Phi(\sqrt{J_F/4}) > \Phi(0) \iff J_F > 0 \iff \mu_+ \neq \mu_- \text{ in Mahalanobis distance.}$$

In the non-Gaussian case, the result follows from the fundamental theorem of linear classifiers: a linear separator with positive margin exists iff the class means are not coincident in the feature space equipped with the Mahalanobis metric induced by Σ. $\square$

4.2 CSS and Mutual Information

Theorem 4.3 (CSS Implies a Mutual Information Lower Bound)**.** *Let Z and Y^c be the random variables corresponding to embeddings and binary concept labels, with balanced class distribution. If* CSS$(c) = \alpha$ *with $\alpha > 1/2$, then:*

$$I(Z; Y^c) \; \geq \; 1 - H_b(1 - \alpha),$$

where $H_b(p) = -p \log_2 p - (1-p) \log_2(1-p)$ is the binary entropy.

Proof. Let $\hat{Y}^c$ be the prediction of the logistic regression probe. By the data processing inequality: $I(Z; Y^c) \geq I(\hat{Y}^c; Y^c)$. The mutual information between a prediction and its binary target is bounded below via Fano's inequality [Cover and Thomas, 2006]: for binary labels,

$$H(Y^c|\hat{Y}^c) \leq H_b(P_e),$$

where $P_e = \mathbb{P}[\hat{Y}^c \neq Y^c]$ is the error rate. Since CSS$(c) = \mathrm{BalAcc}(\hat{Y}^c, Y^c) = 1 - P_e$ for balanced classes, we have $P_e = 1 - \alpha$. Therefore:

$$I(Z; Y^c) \geq I(\hat{Y}^c; Y^c) = H(Y^c) - H(Y^c|\hat{Y}^c) \geq 1 - H_b(1 - \alpha),$$

using $H(Y^c) = 1$ bit for balanced binary labels. $\square$

Corollary 4.4 (CSS Gap Quantifies Information Gain from Pretraining)**.** *Let* $\alpha_p = \mathrm{CSS}_{\mathrm{pretrained}}(c)$ *and* $\alpha_r = \mathrm{CSS}_{\mathrm{random}}(c)$ *denote the pretrained and random-weight CSS for concept* c*. The mutual information gain due to pretraining satisfies:*

$$I(Z_\theta; Y^c) - I(Z_{\theta_0}; Y^c) \;\geq\; H_b(1-\alpha_r) - H_b(1-\alpha_p).$$

Proof. Apply Theorem 4.3 to both the pretrained encoder (with CSS $= \alpha_p$) and the random encoder (with CSS $= \alpha_r$). Subtracting the two lower bounds, and noting that H_b is decreasing on $[0, 1/2]$ (so $\alpha_p > \alpha_r$ implies $H_b(1-\alpha_p) < H_b(1-\alpha_r)$, i.e., the right-hand side is positive), gives the result. □

Example 4.5. For seasonality on Electricity: $\alpha_p = 82.6\%$, $\alpha_r = 64.3\%$. The lower bound gives $\Delta I \geq H_b(0.357) - H_b(0.174) = 0.952 - 0.619 = 0.333$ bits. The pretrained encoder stores at least 0.33 bits more information about seasonality than the random-weight baseline.

4.3 Validity of the Permutation Test

Proposition 4.6 (Permutation Test Exact Under Null)**.** *Under* $H_0 : Z \perp Y^c$ *(independence between embeddings and concept labels), the permutation test p-value*

$$\hat{p} \;=\; \frac{\sum_{j=1}^{n_{\mathrm{perm}}} \mathbb{1}[\mathrm{CSS}_j \geq \mathrm{CSS}_{\mathrm{obs}}] + 1}{n_{\mathrm{perm}} + 1}$$

controls the Type I error at level α *for any* $\alpha \geq 1/(n_{\mathrm{perm}} + 1)$.

Proof. Under H_0, the label vector $\mathbf{y}^c$ is exchangeable: its distribution is invariant under any permutation of its entries. Therefore the observed test statistic $\mathrm{CSS}_{\mathrm{obs}}$ is uniformly distributed among the $n_{\mathrm{perm}} + 1$ values $\{\mathrm{CSS}_0 = \mathrm{CSS}_{\mathrm{obs}}, \mathrm{CSS}_1, \ldots, \mathrm{CSS}_{n_{\mathrm{perm}}}\}$ obtained under permutations (with CSS_0 being the observed value). The probability that $\mathrm{CSS}_{\mathrm{obs}}$ is the maximum of these values is exactly $1/(n_{\mathrm{perm}} + 1)$, so:

$$\mathbb{P}_{H_0}(\hat{p} \leq \alpha) = \mathbb{P}_{H_0}(\mathrm{CSS}_{\mathrm{obs}} \geq \lceil \alpha(n_{\mathrm{perm}} + 1) \rceil\text{-th largest among all } n_{\mathrm{perm}} + 1 \text{ values}) \leq \alpha.$$

□

Remark 4.7. In the main paper, $n_{\mathrm{perm}} = 60$ for in-distribution benchmarks and $n_{\mathrm{perm}} = 20$ for OOD analyses, giving minimum p-values of $1/61 \approx 0.016$ and $1/21 \approx 0.048$ respectively. Reported p-values of 0.0099 correspond to $1/(100+1) \approx 0.0099$ in contexts where 100-resample permutation tests are used (Table **??** caption). Proposition 4.6 guarantees these tests control Type I error exactly.

5 Reliability Geometry: Foundations of PDS and FSS

5.1 k-NN Distance as a Manifold Density Estimator

Theorem 5.1 (k-NN Distance and Local Manifold Density)**.** *Let* $\{z_i\}_{i=1}^{N}$ *be i.i.d. draws from a measure with density* ρ *on a smooth* d_{int}*-dimensional Riemannian manifold*

$(\mathcal{M}, g)$ embedded in $\mathbb{R}^D$. Let $\bar{d}_i^{(k)}$ be the mean distance to the k nearest neighbours of z_i. Then, for fixed k and as $N \to \infty$:

$$\bar{d}_i^{(k)} \;\approx\; c_{d_{\text{int}},k} \cdot \rho(z_i)^{-1/d_{\text{int}}},$$

where $c_{d_{\text{int}},k} = \Big(\Gamma(d_{\text{int}}/2+1)/(\pi^{d_{\text{int}}/2}\,k)\Big)^{1/d_{\text{int}}}$ is a constant depending only on d_{int} and k.

Proof sketch. In a region of constant density ρ on a d_{int}-dimensional manifold, the volume of a geodesic ball of radius r is approximately $\omega_{d_{\text{int}}}\, r^{d_{\text{int}}}$, where $\omega_{d_{\text{int}}} = \pi^{d_{\text{int}}/2}/\Gamma(d_{\text{int}}/2+1)$ is the volume of the unit d_{int}-ball. The expected number of points within radius r of z_i is $N\rho(z_i)\omega_{d_{\text{int}}} r^{d_{\text{int}}}$. Setting this equal to k gives the k-th nearest-neighbour distance: $r_k(z_i) \approx \big(k/(N\rho(z_i)\omega_{d_{\text{int}}})\big)^{1/d_{\text{int}}}$. The mean of the k nearest-neighbour distances scales similarly, yielding the stated formula. A rigorous proof via empirical process theory is given in Penrose [2011]. □

Corollary 5.2 (High $\bar{d}_i$ Signals Low-Density Regions)**.** *Under the conditions of Theorem 5.1, high $\bar{d}_i^{(k)}$ corresponds to low local density $\rho(z_i)$, indicating that z_i lies in a sparse or unexplored region of the manifold.*

5.2 PDS Positivity under Distribution Shift

Theorem 5.3 (PDS Positivity under Monotone Error-Density Relationship)**.** *Assume the forecast error $|e_i|$ is non-decreasing in the Mahalanobis distance $d_{\mathcal{P}}(w_i)$ from window w_i to the pretraining distribution $\mathcal{P}_{\text{train}}$, and that $\bar{d}_i$ is non-decreasing in $d_{\mathcal{P}}(w_i)$. Then the Spearman correlation $\rho_s^{\text{PDS}} = \text{Spearman}(\bar{d}_i, |e_i|) \geq 0$.*

Proof. Both $\bar{d}_i$ and $|e_i|$ are non-decreasing functions of the same latent variable $d_{\mathcal{P}}(w_i)$. Therefore their ranks are non-decreasingly related: if $d_{\mathcal{P}}(w_i) \leq d_{\mathcal{P}}(w_j)$, then $\bar{d}_i \leq \bar{d}_j$ and $|e_i| \leq |e_j|$. Concordant rank pairs outnumber discordant pairs in expectation, so Kendall's $\tau \geq 0$ and equivalently Spearman's $\rho_s \geq 0$. Strict positivity follows whenever $d_{\mathcal{P}}$ has non-degenerate variation across $\{w_i\}$. □

Remark 5.4. The assumption that $\bar{d}_i$ is non-decreasing in $d_{\mathcal{P}}(w_i)$ (distribution shift) is justified by Theorem 5.1: windows far from the pretraining distribution appear in low-density regions of $\mathcal{M}_\theta$, hence have large $\bar{d}_i$. The assumption that $|e_i|$ is non-decreasing in distribution shift is standard in learning theory (see Theorem 6.2 below).

5.3 Local vs. Global Information: PDS versus FSS

Theorem 5.5 (PDS Can Strictly Exceed FSS)**.** *Let $\mathcal{M}_\theta$ have at least two connected components with substantially different densities: a dense cluster A (low density) and a sparse cluster B (high density) with mean forecast errors $\bar{e}_A < \bar{e}_B$. Assume A and B are close globally but well-separated locally. Then there exist distributions for which $\rho_s^{\text{PDS}} \gg \rho_s^{\text{FSS}}$.*

Proof. *Construction.* Let $A = \{z_i\}_{i=1}^{n/2}$ with $z_i \sim \mathcal{N}(0, \sigma_A^2 I)$ (dense, σ_A small, low error), and $B = \{z_i\}_{i=n/2+1}^{n}$ with $z_i \sim \mathcal{N}(\mu, \sigma_B^2 I)$ (sparse, $\sigma_B \gg \sigma_A$, high error), where $\|\mu\| \approx 0$ (the clusters overlap globally).

FSS analysis. Global pairwise distances $\|z_i - z_j\|_2$ are similar whether (i, j) is an A-A, A-B, or B-B pair (since $\|\mu\| \approx 0$). Forecast error differences $|e_i - e_j|$ are also similar on average across pair types. Therefore $\rho_s^{\mathrm{FSS}} \approx 0$.

PDS analysis. For $z_i \in A$: $\bar{d}_i^{(k)} \approx c\sigma_A$ (small, low error). For $z_i \in B$: $\bar{d}_i^{(k)} \approx c\sigma_B$ (large, high error). Since $\sigma_B \gg \sigma_A$ and $\bar{e}_B \gg \bar{e}_A$, the rank correlation ρ_s^{PDS} is high.

Since σ_B/σ_A can be made arbitrarily large, ρ_s^{PDS} can be made arbitrarily close to 1 while $\rho_s^{\mathrm{FSS}} \approx 0$. $\square$

6 Generalization Bounds via Manifold Complexity

6.1 Covering Numbers of the TRM

Theorem 6.1 (Covering Number Bound for the TRM)**.** *Under Assumptions 1.2 and 1.3, the ϵ-covering number of $\mathcal{M}_\theta$ with respect to the Euclidean metric satisfies:*

$$\mathcal{N}(\mathcal{M}_\theta, \|\cdot\|_2, \epsilon) \le \left(\frac{2\, L_\theta\, R_w}{\epsilon}\right)^{d_{\mathrm{int}}},$$

where $R_w = \mathrm{diam}(\mathrm{supp}(\mathcal{P}))/2$ is the radius of the input support and $d_{\mathrm{int}} = \min(L, D)$.

Proof. The input support $\mathrm{supp}(\mathcal{P}) \subset \mathbb{R}^L$ can be covered by at most $(2R_w/\delta)^L$ Euclidean balls of radius δ. The encoder f_θ is L_θ-Lipschitz, so it maps each ball of radius δ to a set of diameter at most $2L_\theta\delta$ in $\mathbb{R}^D$. Setting $2L_\theta\delta = \epsilon$, i.e., $\delta = \epsilon/(2L_\theta)$, gives a cover of $\mathcal{M}_\theta$ by at most $(2R_w/\delta)^L = (4R_w L_\theta/\epsilon)^L$ sets of diameter ϵ. The intrinsic dimension $d_{\mathrm{int}} \le L$ (from Theorem 1.5), so the bound $(2L_\theta R_w/\epsilon)^{d_{\mathrm{int}}}$ holds when the manifold has lower intrinsic dimension than the ambient input space. $\square$

6.2 Generalization Bound via Manifold Structure

Theorem 6.2 (Generalization Bound from Manifold Complexity)**.** *Let $\mathcal{F}$ be a class of predictors $\hat{e} : \mathcal{M}_\theta \to \mathbb{R}$ with $|\hat{e}(z)| \le M$ and $\mathrm{Lip}(\hat{e}) \le G$. For N i.i.d. samples from $\mathcal{P}$, with probability at least $1 - \delta$:*

$$\sup_{\hat{e}\in\mathcal{F}} \left|L(\hat{e}) - \hat{L}_N(\hat{e})\right| \le 2GM\sqrt{\frac{d_{\mathrm{int}}\, \log(2L_\theta R_w/\epsilon)}{N}} + 3M\sqrt{\frac{\log(2/\delta)}{2N}},$$

where $L(\hat{e}) = \mathbb{E}[\ell(\hat{e}(z), e)]$ is the true risk and $\hat{L}_N$ is the empirical risk.

Proof. We apply the standard Rademacher complexity generalization bound:

$$\sup_{\hat{e}} |L - \hat{L}_N| \le 2\mathfrak{R}_N(\mathcal{F}) + 3M\sqrt{\tfrac{\log(2/\delta)}{2N}},$$

where $\mathfrak{R}_N(\mathcal{F})$ is the empirical Rademacher complexity. For a class of G-Lipschitz functions on a set with covering number $\mathcal{N}(\mathcal{M}_\theta, \epsilon)$, the Rademacher complexity is bounded by

$$\mathfrak{R}_N(\mathcal{F}) \le GM\sqrt{\frac{\log \mathcal{N}(\mathcal{M}_\theta, \epsilon)}{N}} + \epsilon G \le GM\sqrt{\frac{d_{\text{int}}\, \log(2L_\theta R_w/\epsilon)}{N}} + \epsilon G,$$

using Theorem 6.1 for the covering number. Optimising over ϵ gives the stated bound. □

Corollary 6.3 (Low Intrinsic Dimension Tightens Generalization)**.** *Under the conditions of Theorem 6.2, if encoder f_{θ_1} has intrinsic dimension $d_{\text{int}}^{(1)}$ and encoder f_{θ_2} has $d_{\text{int}}^{(2)} > d_{\text{int}}^{(1)}$, and both achieve the same empirical risk, then f_{θ_1} has a tighter generalization bound:*

$$L(f_{\theta_1}) - \hat{L}_N(f_{\theta_1}) \;\le\; \sqrt{\frac{d_{\text{int}}^{(1)}}{d_{\text{int}}^{(2)}}}\,\big(L(f_{\theta_2}) - \hat{L}_N(f_{\theta_2})\big) + O(N^{-1/2}).$$

Proof. The generalization gap scales as $O(\sqrt{d_{\text{int}}/N})$ by Theorem 6.2. Taking the ratio gives the bound directly. □

Remark 6.4. Corollary 6.3 provides a theoretical basis for the observed 3.5× LTC step gap between the pretrained TSFM and the supervised Transformer: the pretrained encoder produces a lower-dimensional manifold (lower d_{int}), which by Corollary 6.3 implies a tighter generalization bound on unseen windows—consistent with the superior zero-shot performance of TSFMs.

6.3 Distribution Shift and OOD Error Growth

Theorem 6.5 (OOD Error Bound via Manifold Distance)**.** *Let $\mathcal{P}_{\text{train}}$ and $\mathcal{P}_{\text{test}}$ be the training and test window distributions. Let $d_{\mathcal{M}}(\mathcal{P}_{\text{train}}, \mathcal{P}_{\text{test}})$ denote the discrepancy between the induced measures on $\mathcal{M}_\theta$ (e.g., maximum mean discrepancy or Wasserstein distance). For any predictor $\hat{e}$ trained to minimise risk under $\mathcal{P}_{\text{train}}$:*

$$L_{\text{test}}(\hat{e}) \;\le\; L_{\text{train}}(\hat{e}) + C\,G\,d_{\mathcal{M}}(\mathcal{P}_{\text{train}}, \mathcal{P}_{\text{test}}),$$

where $G = \mathrm{Lip}(\hat{e})$ and $C > 0$ is an absolute constant.

Proof. By the definition of Wasserstein-1 distance W_1:

$$|L_{\text{test}}(\hat{e}) - L_{\text{train}}(\hat{e})| = |\mathbb{E}_{\text{test}}[\ell] - \mathbb{E}_{\text{train}}[\ell]| \le G\,W_1(\mathcal{P}_{\text{test}}, \mathcal{P}_{\text{train}}),$$

since a G-Lipschitz loss function is a valid test function in the Kantorovich dual representation of W_1. Identifying $d_{\mathcal{M}} = W_1$ completes the proof; other integral probability metrics (MMD) give analogous bounds with different constants. □

Corollary 6.6 (PDS Collapse on OOD Data)**.** *If $d_{\mathcal{M}}(\mathcal{P}_{\text{train}}, \mathcal{P}_{\text{test}})$ is large (deep OOD data), then the embedding density $\rho(z)$ on $\mathcal{M}_\theta$ is low for $z \in f_\theta(\text{supp}(\mathcal{P}_{\text{test}}))$. Consequently, the rank correlation between $\bar{d}_i$ and $|e_i|$ is disrupted because:*

(i) all OOD embeddings have large $\bar{d}_i$ (uniformly low density), removing variance in $\bar{d}_i$ needed for Spearman correlation;

(ii) the relationship between $\bar{d}_i$ and $|e_i|$ is no longer monotone, since $|e_i|$ depends on OOD-specific factors unrelated to manifold structure.

Remark 6.7. Corollary 6.6 provides the theoretical explanation for the empirical finding that $\rho_s^{\text{PDS}} \approx 0$ on CSI300 and CSI800 equity data while ρ_s^{PDS} is significant on Electricity and Traffic. Chinese equity indices lie far outside the support of any known TSFM pretraining distribution (financial daily data at daily resolution), satisfying the conditions of the corollary exactly. In contrast, LTC step and CSS remain above random-weight baselines on OOD data (Theorem 2.6 and Corollary 4.4 hold regardless of distribution shift), consistent with the empirical observation that trajectory structure and pattern encoding transfer to equity data while the reliability signal does not.

7 Connections to Geometric Deep Learning

7.1 Equivariance, Invariance, and Temporal Structure

The geometric deep learning framework of Bronstein et al. [2021] unifies neural architectures through the symmetry groups they respect. For time series, the relevant group is the group of time shifts and scalings; we discuss how LGF metrics relate to this structure.

Definition 7.1 (Temporal Shift Equivariance)**.** An encoder f_θ is *shift-equivariant* with respect to translations $\tau_s : w(t) \mapsto w(t-s)$ if:

$$f_\theta(\tau_s(w)) \;=\; T_s(f_\theta(w))$$

for some action T_s on $\mathbb{R}^D$.

Proposition 7.2 (Shift Equivariance Implies Bounded LTC_{dir})**.** *If f_θ is shift-equivariant and T_s is a rotation group action, then LTC_{dir} is determined by the angular velocity of the orbit of z_i under T_s, and is bounded away from -1.*

Proof. If T_s is an isometric group action (rotation), then $\|f_\theta(w_{t+1}) - f_\theta(w_t)\|_2 = \|T_s z_t - z_t\|_2$, which depends only on the angle of rotation θ_s. The cosine similarity between consecutive steps is $\cos(\angle(\delta_t, \delta_{t+1})) = \cos(2\theta_s)$, which is constant and bounded below by $-1 + 2\sin^2\theta_s > -1$ for non-trivial rotation ($\theta_s \neq \pi$). Averaging over t gives $\text{LTC}_{\text{dir}} > -1$. □

7.2 The Manifold Hypothesis and Temporal Primitives

The *manifold hypothesis*—that natural high-dimensional data lies on a low-dimensional manifold—provides the foundational justification for using distance-based metrics (LTC, PDS, FSS) on embedding spaces.

Theorem 7.3 (Temporal Primitives Correspond to Submanifolds). *Let $c \in \{\text{trend}, \text{season}, \text{vol}\}$ be a binary temporal concept. If the encoder f_θ encodes concept c (i.e.,* $\mathrm{CSS}(c) > 1/2$*), then there exist two connected open sets $\mathcal{M}_c^+, \mathcal{M}_c^- \subset \mathcal{M}_\theta$ corresponding to the two classes of c, separated by a hyperplane in $\mathbb{R}^D$.*

Proof. $\mathrm{CSS}(c) > 1/2$ implies (by Theorem 4.2) that a linear hyperplane $H = \{z : w^\top z = b\}$ achieves better-than-chance separation. The two classes are therefore separated by H, so: $\mathcal{M}_c^+ = \mathcal{M}_\theta \cap \{z : w^\top z > b\}$ and $\mathcal{M}_c^- = \mathcal{M}_\theta \cap \{z : w^\top z < b\}$ are disjoint open (relative to $\mathcal{M}_\theta$) submanifolds. □

Remark 7.4. Theorem 7.3 formalises the geometric picture behind the CSS results: for seasonality and volatility (both CSS $> 50\%$), the TRM decomposes into two separated regions corresponding to the two states of each concept. For trend direction (CSS $\approx 50\%$ on most datasets), no such hyperplane exists in the frozen embedding space—a consequence of window-level normalisation removing the global offset information that would distinguish trend directions.

7.3 Geometric Stability and Robust Representation Learning

We conclude with a theorem connecting the stability properties measured by TSI and GSP to the broader goal of robust representation learning.

Theorem 7.5 (Stability Implies Bounded Rademacher Complexity). *Let $h : \mathcal{M}_\theta \to [0, 1]$ be a predictor class defined by linear classifiers on frozen embeddings. If $\mathrm{TSI}_\mathcal{P} \geq 1 - \kappa$ for all perturbations $\mathcal{P}$ with $\|\mathcal{P}(w) - w\|_2 \leq \epsilon$, then the Rademacher complexity of h satisfies:*

$$\mathfrak{R}_N(h) \;\leq\; \frac{2B}{\sqrt{N}} + C\kappa^{1/2},$$

where B bounds the norm of the linear classifier weights and C is a constant depending on R and $\bar{d}$.

Proof. The Rademacher complexity of linear classifiers on a compact set is $O(B/\sqrt{N})$ by standard results (see, e.g., Shalev-Shwartz and Ben-David 2014). The perturbation stability ($\mathrm{TSI} \geq 1 - \kappa$) implies that the effective diameter of the embedding set—the maximum distance between the clean and perturbed version of the same input—is at most $2R\kappa^{1/2}$. This adds an additive perturbation term $O(\kappa^{1/2})$ to the Rademacher complexity via the standard stability-to-generalization argument (cf. Hardt et al. 2016). □

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